



The Double-Edged Sword of Digital Technologies

Come la AI influenza il carico cognitivo e il benessere

Stefano Lambiase

Assistant Professor, Aalborg University, Copenhagen

Il 73% dei giovani adulti
soffre di solitudine



M E N T A L

WELN
S T R E S S

La tecnologia ha reso la gestione del nostro benessere mentale un'attività socio-tecnica complessa.



Sovraccarico Informativo

Il sovraccarico informativo si verifica quando si è esposti a una quantità di informazioni superiore alla capacità di elaborarle, causando difficoltà nel prendere decisioni, stress, ansia e ridotta produttività.



Tecnodipendenza

La tecnodipendenza è un disturbo comportamentale caratterizzato da un uso eccessivo e compulsivo di tecnologie digitali, che può portare a isolamento sociale, problemi di salute fisica e mentale e una riduzione delle attività nella vita reale.



State sprecando il vostro **tempo**





ChatGPT
came into
our lives





Affidamento Eccessivo

Affidarsi a strumenti tecnologici che utilizzano l'intelligenza artificiale generativa (come ChatGPT) per gestire momenti di stress e tristezza, al punto da considerare lo strumento un confidente migliore delle persone reali.



DEGENERATION



DEGENERATION







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Copenhagen
Denmark



Assistant Professor in Software Engineering

Un appassionato (e abile) giocatore di videogiochi che possiede solo camicie e nutre un fascino particolare per lo stile classico da gentiluomo inglese (o da boscaiolo).

Per qualche misteriosa ragione, ama anche la ricerca, la gestione di progetti e cucinare lasagne. Non sopporta la vista di tabelle mal formattate in LaTeX.

Research Topics

- Human Aspects of Software Engineering
- Generative AI for Software Engineering
- Video Game Development and Engineering

Objective



Cognitive Sustainability

L'obiettivo di questa chiacchierata è quello di discutere di **cognitive sustainability**, cioè l'abilità di mantenere equilibrio, attenzione, memoria e agency in un mondo dominato da Generative AI.

Per farlo, sottolineeremo sia gli aspetti positivi che negativi di queste tecnologie nelle nostre vite.



Lati Negativi



1 — L'AI può renderci più pigri e meno capaci



Cognitive Decline

L'uso prolungato dell'IA generativa può ridurre la vostra capacità di impegnarvi nel pensare e nel giudizio critico!





Usare l'AI senza
pensare equivale
a contrarre un
debito con la
vostra mente che
potreste non
essere mai capaci
di ripagare.

Un esempio...





2 — L'AI può umentare il nostro stress e deprimerci



L'AI può portare a dipendenza
psicologica e stress emotivo...



L'AI può portare a dipendenza
psicologica e stress emotivo...

... anche quando non usata
con il preciso obiettivo di avere
un supporto emotivo...

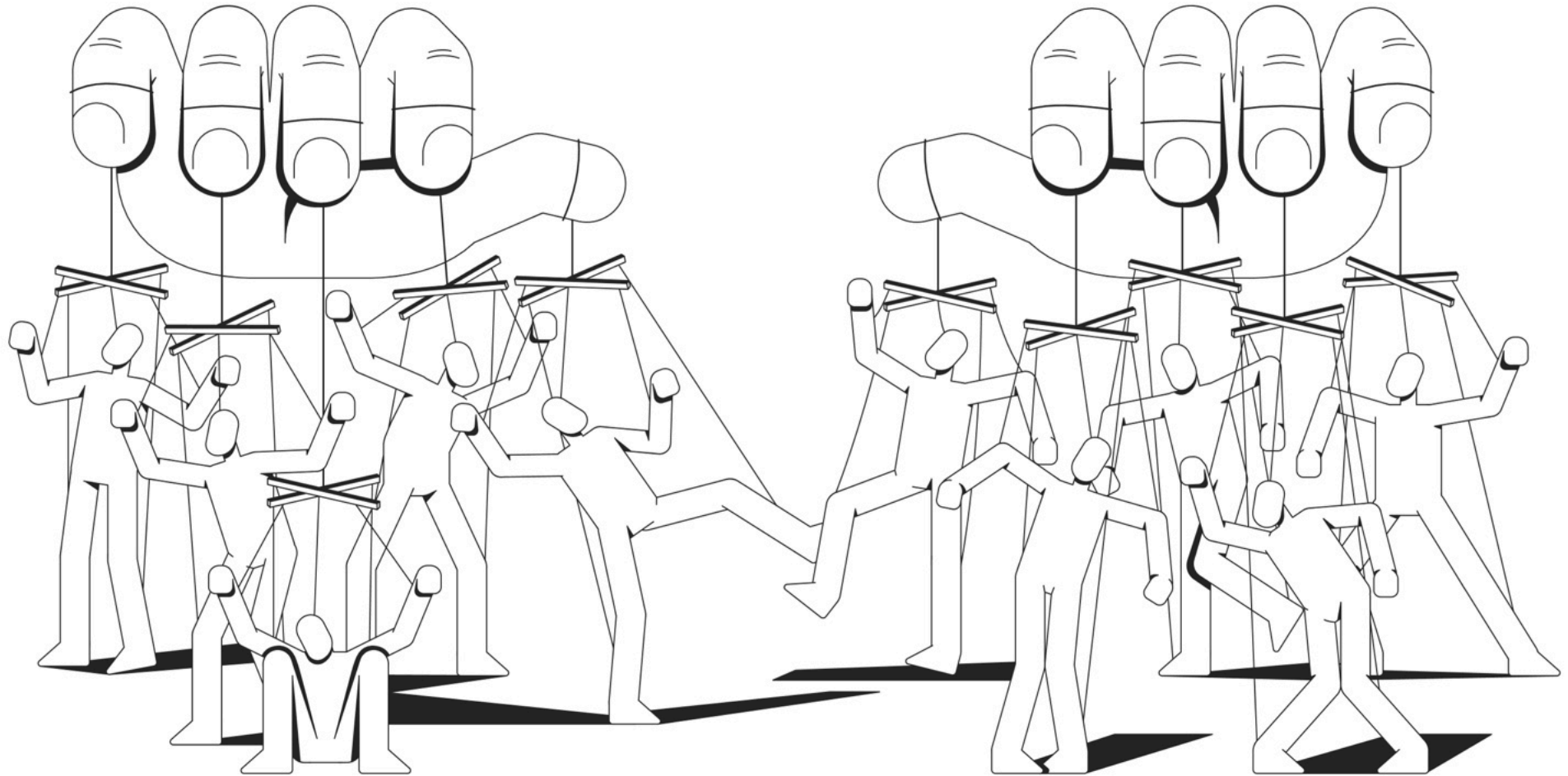




WHY?



Mancanza di Controllo



2 unkind. 10
Peter was fat.

meaning /'mi:nɪŋ/
what something means
word has two different
senses /'mi:nɪz/ no

**Evitare di
ascoltare le
nostre
emozioni**



3 — L'AI può renderci persone peggiori





AI HALLUCINATION

> DEFINITION

When AI systems generate false or nonsensical information that appears plausible but is not based on training data



BIAS

Unfair

Beliefs

Prejudice

Preferences

Stereotypes







La solitudine può
portarci a usare l'AI
generativa...

... ma l'uso di AI
generative può farci
sviluppare atteggiamenti
che portano a isolarci...





A close-up shot of a man with dark hair and a light beard, looking slightly upwards and to the left with a pensive or questioning expression. He is wearing a dark suit jacket over a light blue shirt. The background is blurred, showing what appears to be a window with horizontal blinds.

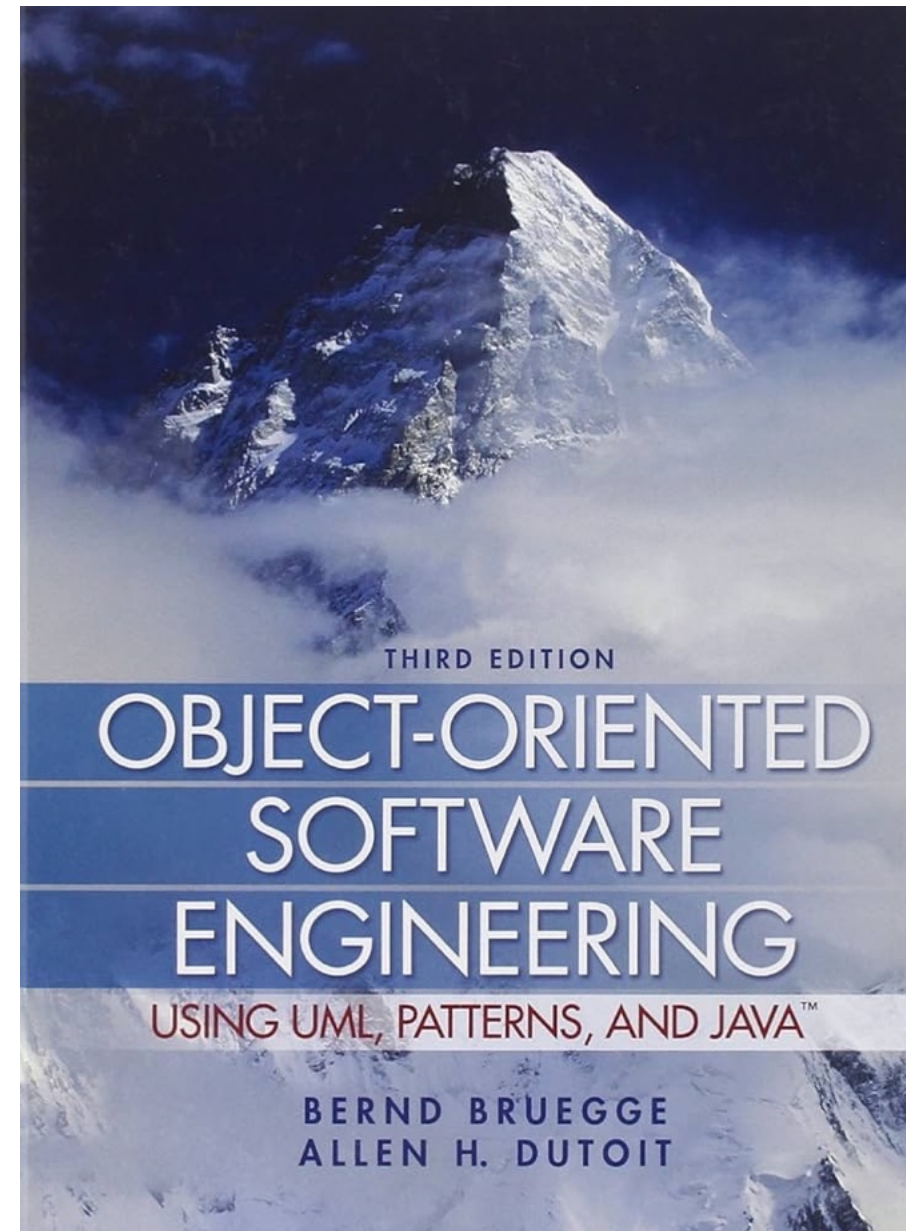
Quindi, siamo destinati a soffrire?



Lati Positivi

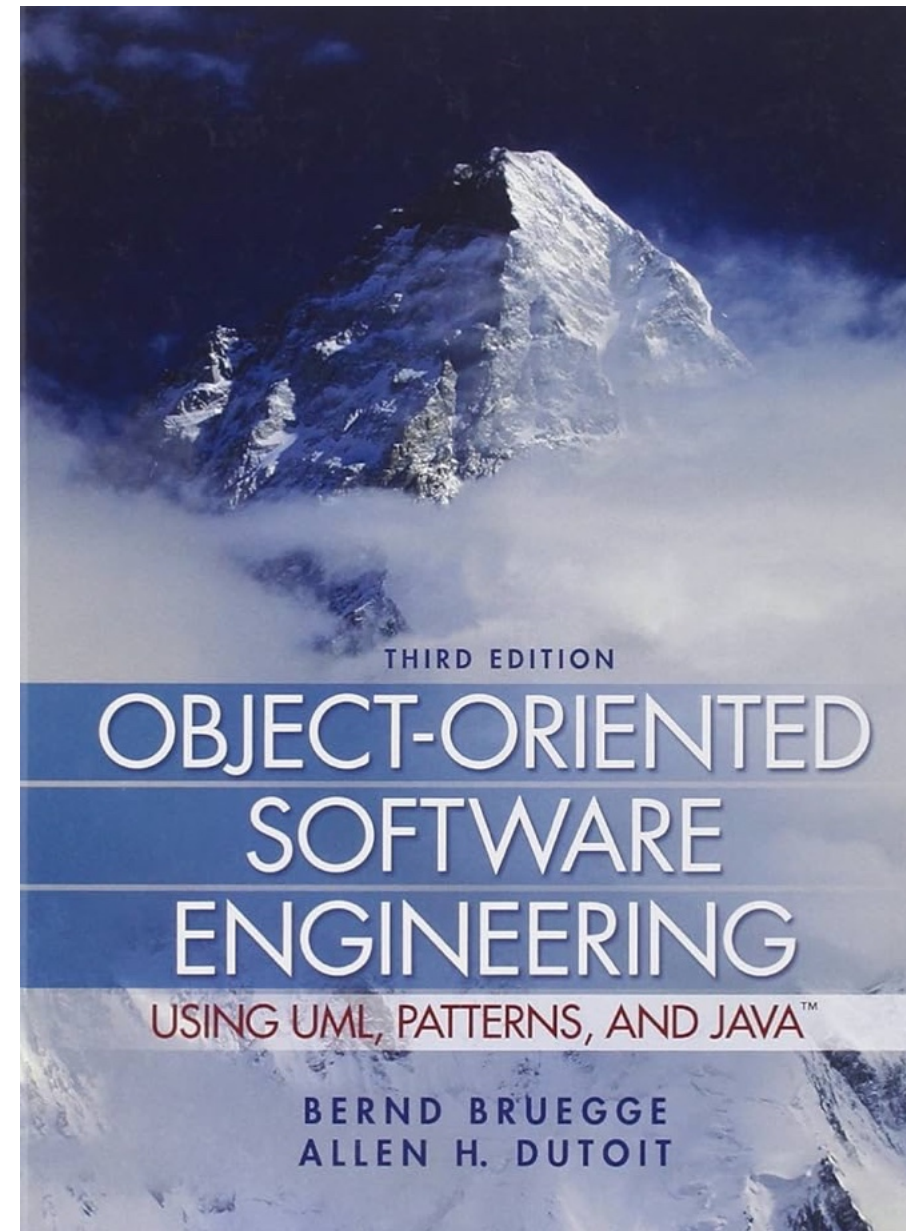


Il libro proviene dal corso di ingegneria del software della Triennale. È un libro eccellente, ma ha un piccolo problema:



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È GRANDE







WIKIPEDIA
The Free Encyclopedia



Siamo sommersi di
informazioni



Just-In-Time Test Smell Detection and Refactoring: The DARTS Project

Stefano Lambiase, Andrea Cupito, Fabiano Pecorelli, Andrea De Lucia, Fabio Palomba
SeSa Lab - University of Salerno, Italy

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ABSTRACT

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CCS CONCEPTS

• Software and its tools.

KEYWORDS

Test Smells, Refactoring, ACM Reference Format: Stefano Lambiase, Andrea Cupito, Fabiano Pecorelli, Andrea De Lucia, Fabio Palomba, 2018. Just-In-Time DARTS Project. In *Proceedings of the ACM Conference on Program Comprehension*. <https://doi.org/10.1145/3112122>

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Alexandre a.serehin
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ABSTRACT

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"There and Back Again?" On the Influence of Software Community Dispersion Over Productivity

Stefano Lambiase¹, Gemma Catolino¹, Fabiano Pecorelli¹, Damiano A. Tamburri¹, Fabio Palomba¹,
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Abstract—Estimating and understanding productivity still represents a crucial task for researchers and practitioners. Researchers spent significant effort identifying the factors that influence software developers' productivity, providing several approaches for analyzing and predicting such a metric. Although different works focused on evaluating the impact of human factors on productivity, little is known about the influence of cultural/geographical diversity in software development communities. Indeed, in previous studies, researchers treated cultural aspects like an abstract concept without providing a quantitative representation. This work provides an empirical assessment of the relationship between cultural and geographical dispersion of a development community—namely, how diverse a community is in terms of cultural attitudes and geographical collocation of the members who belong to it—and its productivity. To reach our aim, we built a statistical model that contained product and socio-technical factors as independent variables to assess the correlation with productivity, i.e., the number of commits performed in a given time. Then, we ran our model considering data of 25 open-source communities on GitHub. Results of our study indicate that cultural and geographical dispersion impact productivity, thus encouraging managers and practitioners to consider such aspects during all the phases of the software development lifecycle.

Index Terms—Global Software Engineering; Cultural Dispersion; Geographical Dispersion; Productivity; Software Organizational Structures; Empirical Studies.

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Nowadays, software development is, even more, a globally distributed activity [1], [2], thus involving stakeholders from different places globally and, consequently, different cultures and backgrounds. Researchers in the context of *Global Software Engineering* (GSE) [3], [4]—i.e., the set of practices and guidelines aimed at managing software distributed teams—demonstrated that such heterogeneity in development communities leads to positive behavior [1], [5] but, if not well managed, it can lead to some issues, e.g., difficulties in communication and misunderstandings between team members [4], [6], [7]. Furthermore, aspects related to developers' communication and interaction have been discovered to be related to the productivity of a software community [8]–[13].

Among the various aspects taken into account by GSE researchers, culture—the way people act and think [14]—is defined as one of the most complex factors with respect to several software metrics [15]–[18]. For defining culture, researchers started using the *Hofstede's Framework* [19], [20]

to represent the culture of individuals. The framework consists of six behavioral characteristics—which can assume values from zero to one hundred—that can culturally characterize individuals according to their home country. The extensive research on productivity mentioned above showed how essential human aspects are in treating this complex topic. However, previous works did not focus the attention on cultural aspects from a concrete perspective.

With this study, we aimed to address the limitations mentioned above by conducting a quantitative empirical study on the relation between the cultural and geographical dispersion of a community—i.e., the degree to which a community is formed by individuals growing up in and coming from different places globally—and its productivity.

To operationalize the cultural dispersion of a development community, we use the well-known Hofstede Framework [19], [20], already used in previous works in SE [21]–[24]. As for the geographical dispersion, we used the spherical distance between the community's members. Afterward, we built a statistical regression model to assess the relationship between the two dispersion metrics and productivity, conducting the study on 25 open-source communities on GitHub.

The key results of our study indicate that dispersion metrics influence developers' productivity both positively and negatively. For example, the presence of people coming from a more rigid culture in contrast to others who are more flexible could reduce the team's work rate. In addition, having team members who agree on the best way to work together—whether closely or in isolation—can positively impact productivity. To sum up, our article provides two contributions:

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These results shed light on an important aspect often underestimated by managers. Indeed, the health status of a community—their behaviors and diversity—needs to be carefully monitored when dealing with software development since it can affect the quantity (productivity) and quality of what produces.

Regarding the structure of the paper, Section II describes the existing literature related to cultural factors in software

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“Se non c’è
l’observer, allora
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1 – L’AI può ridurre il carico cognitivo, riassumendo le informazioni

“German c’è l’oboe, allora non è un MVC”

L'IA generativa ha il potenziale di ridurre il carico cognitivo, aiutandoci a trovare le informazioni più rapidamente e a concentrarci maggiormente su ciò che conta davvero.



*“Eh, ma allora le
allucinazioni?!?”*



*“Eh, ma allora le
allucinazioni?!?”*

Pensiero Critico





Semantic **Scholar**

2 — L'AI può aiutarci nella prima risposta alla sofferenza mentale



Ascolto





Giudizio

A person wearing a dark blue hoodie with the hood pulled up, completely obscuring their face. They are sitting at a desk, and a silver laptop is open in front of them. The background is a dark, textured blue-grey. The word "Anonimato" is written in a black, sans-serif font across the center of the laptop screen.

Anonimato

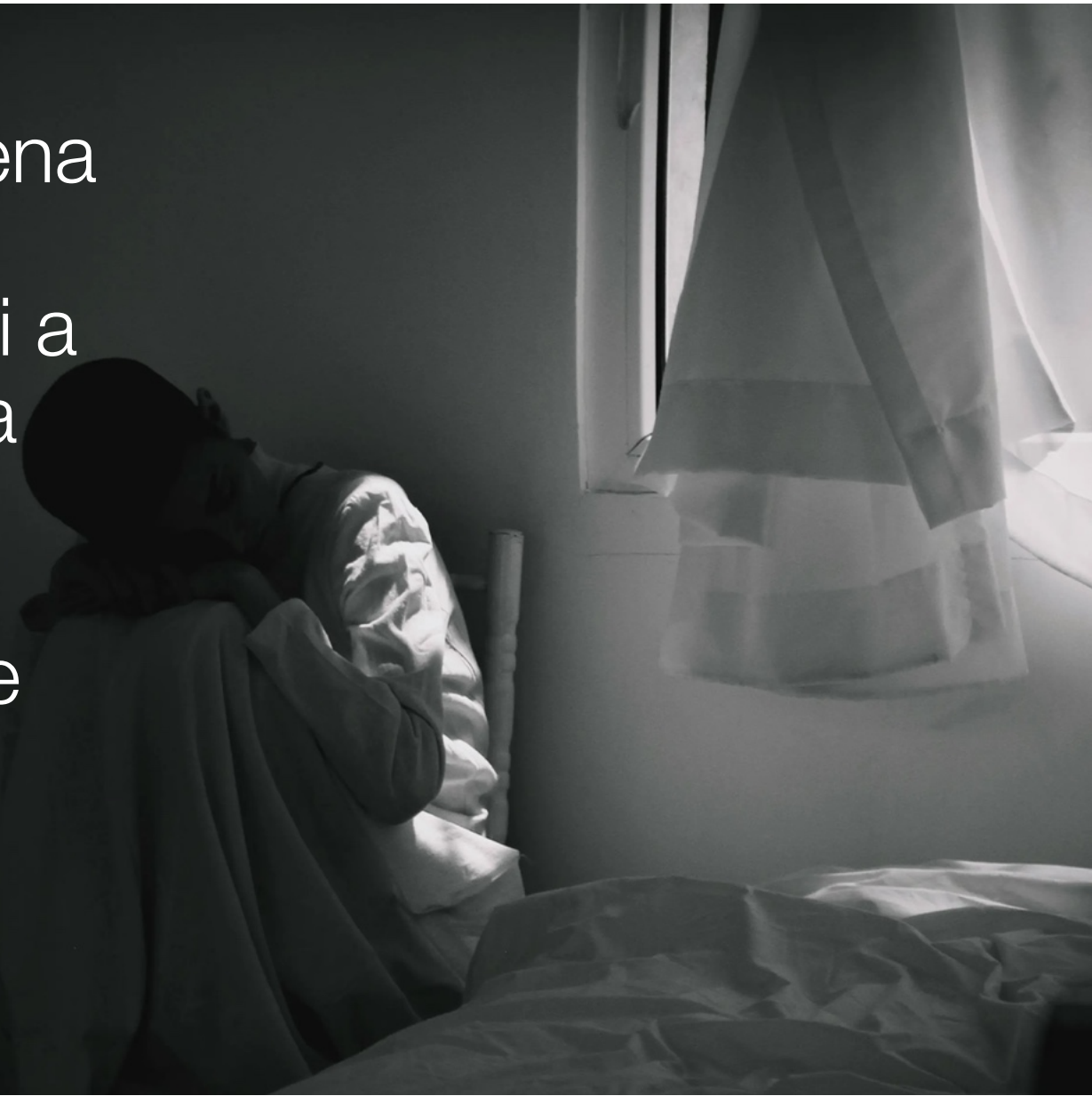
Ascolto

Purtroppo, questi strumenti sono spesso percepiti come meno critici e più empatici delle persone. Inoltre, sono disponibili 24 ore su 24 e per questo motivo possono essere percepiti come un sostegno alla solitudine.



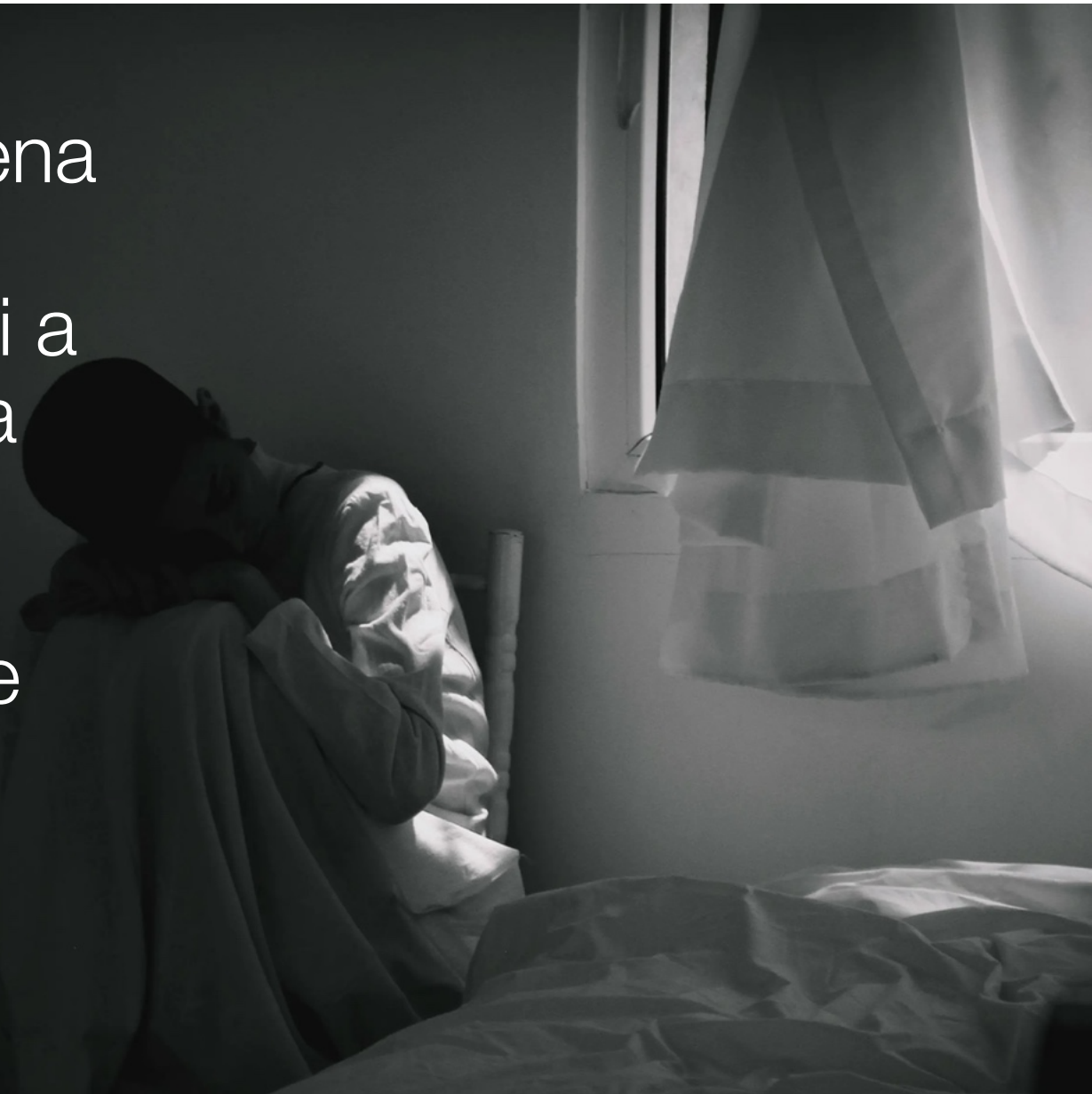
Immagina di aver appena subito un lutto inaspettato e di trovarti a casa, da solo, nella tua camera da letto.

È tardi, tutti dormono e non hai nessuno a cui chiedere compagnia o aiuto.



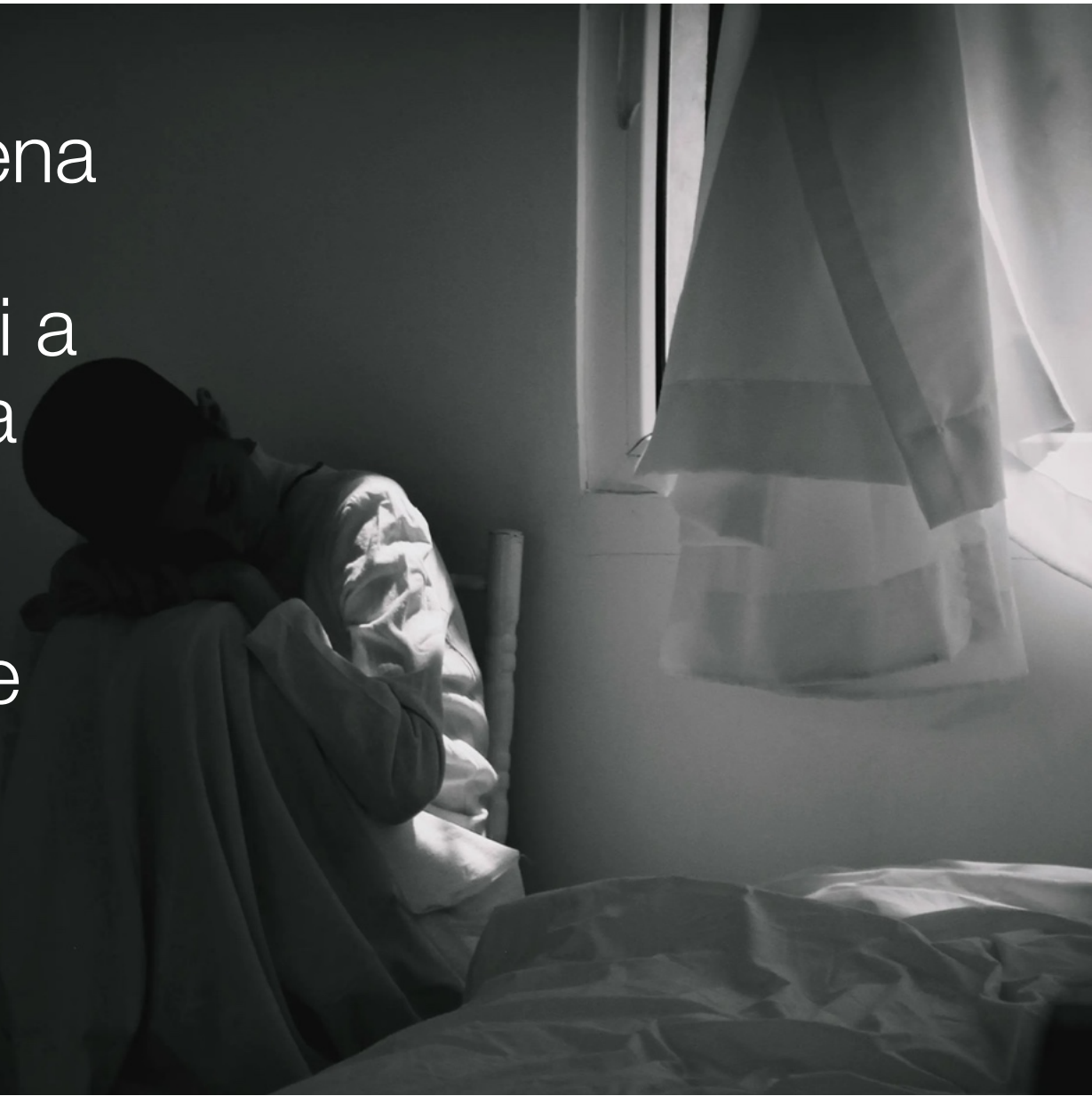
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Immagina di aver appena subito un lutto inaspettato e di trovarti a casa, **da solo**, nella tua camera da letto.

È tardi, tutti dormono e **non hai nessuno** a cui chiedere compagnia o aiuto.





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3 — L'AI generativa può essere usata per stimolare la creatività



L'AI generativa può essere usata per stimolare la creatività

Anche nel contesto dell'ingegneria del software

The Impact of Generative AI on Creativity in Software Development: A Research Agenda

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As GenAI becomes embedded in developer toolchains and practices, and routine code is increasingly generated, human creativity will be increasingly important for generating competitive advantage. This article uses the McLuhan tetrad alongside scenarios of how GenAI may disrupt software development more broadly, to identify potential impacts GenAI may have on creativity within software development. The impacts are discussed along with a future research agenda comprising five connected themes that consider how individual capabilities, team capabilities, the product, unintended consequences, and society can be affected.

CCS Concepts: • **Software and its engineering** → **Software development techniques**; • **Human-centered computing** → **Collaborative and social computing**.

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**The influence of generative
artificial intelligence on creative
cognition of design students: a
chain mediation model of
self-efficacy and anxiety**

Younjung Hwang¹ and Yi Wu^{2*}

¹School of Design, Hunan University, Changsha, China, ²School of International Communication and
Arts, Hainan University, Haikou, China

Introduction: This study investigated the role of generative Artificial Intelligence (AI) in enhancing the creative cognition of design students, examining the mediating effects of self-efficacy and anxiety reduction.

Methods: A quantitative approach was employed, collecting data through online surveys from 121 design students at universities in southern China. The study utilized scales for AI knowledge and perception, self-efficacy, anxiety, and creative cognition, adapted from previous studies and evaluated on 5-point Likert scales. Data analysis was conducted using SPSS 24.0 for exploratory factor analysis and PROCESS v3.5 for mediation analysis.

Results: The findings confirmed that AI positively impacted students' innovative thinking ($\beta^* = 0.610$, $p^* < 0.001$). Self-efficacy (standardized $\beta^* = 0.256$, 95% CI [0.140, 0.418], $p^* < 0.001$) and anxiety reduction (standardized $\beta^* = 0.093$, 95% CI [0.018, 0.195], $p^* < 0.05$) positively mediated the relationship between generative AI and creative cognition. Additionally, a serial mediation effect through self-efficacy and anxiety reduction was observed (standardized $\beta^* = 0.053$, 95% CI [0.012, 0.114], $p^* < 0.05$).

Discussion: Our empirical analysis demonstrates that AI positively affects design students' innovative thinking, with self-efficacy and anxiety reduction serving as significant mediators. These findings provide valuable insights for educators and policymakers, suggesting that AI-integrated design curricula can significantly foster creative cognition, promote academic achievement, and enhance designer capabilities. Understanding AI's impact on students' creative processes is crucial for developing effective teaching strategies in today's evolving educational landscape.

KEYWORDS
generative AI, creative cognition, self-efficacy, anxiety, design education

1 Introduction

In recent years, Artificial Intelligence (AI) has emerged as a transformative force across various domains, including education and creative industries. AI refers to computer systems capable of performing complex tasks traditionally associated with human cognition, such as reasoning, decision-making, and problem-solving (Russell and Norvig, 2020). Generative AI, a subset of AI, is designed to analyze patterns in large datasets and use these patterns to generate outcomes based on user requirements (Mandapuram et al., 2018; Liang et al., 2023). As this technology gains prominence in creative disciplines, its integration in design education

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questi tool
hanno il
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**Dall'altro, essi
hanno il
potenziale di
rendere la
nostra vita più
felice**

Cosa diavolo
vuoi che
faccia?!?





Psicologico

Generative AI per
supporto psicologico

L'AI può render
persone peggio

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MEMORY

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informazioni

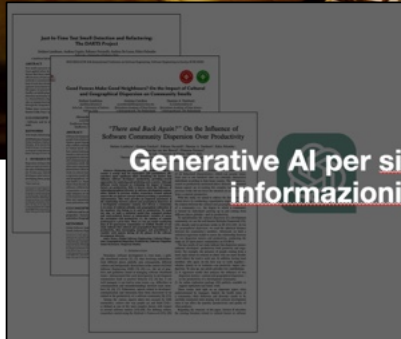
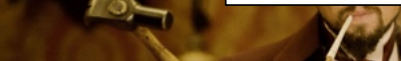
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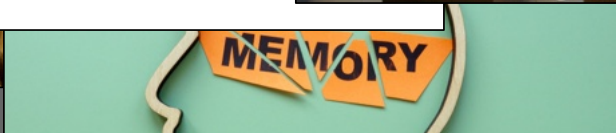
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Troppe informazioni.
Mi fai un riassunto?

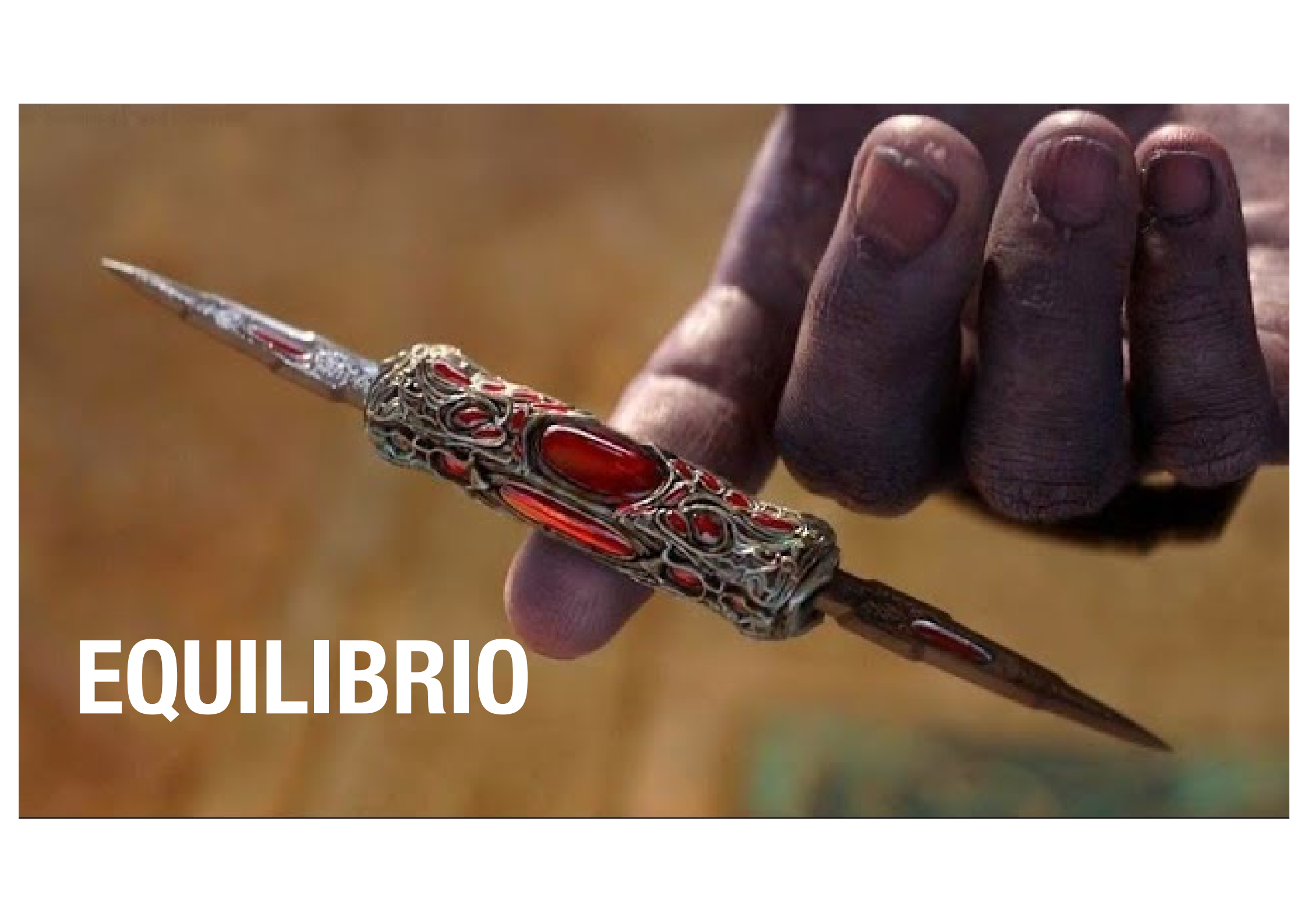


Troppe informazioni.
Mi fai un riassunto?



Troppe informazioni.
Mi fai un riassunto?



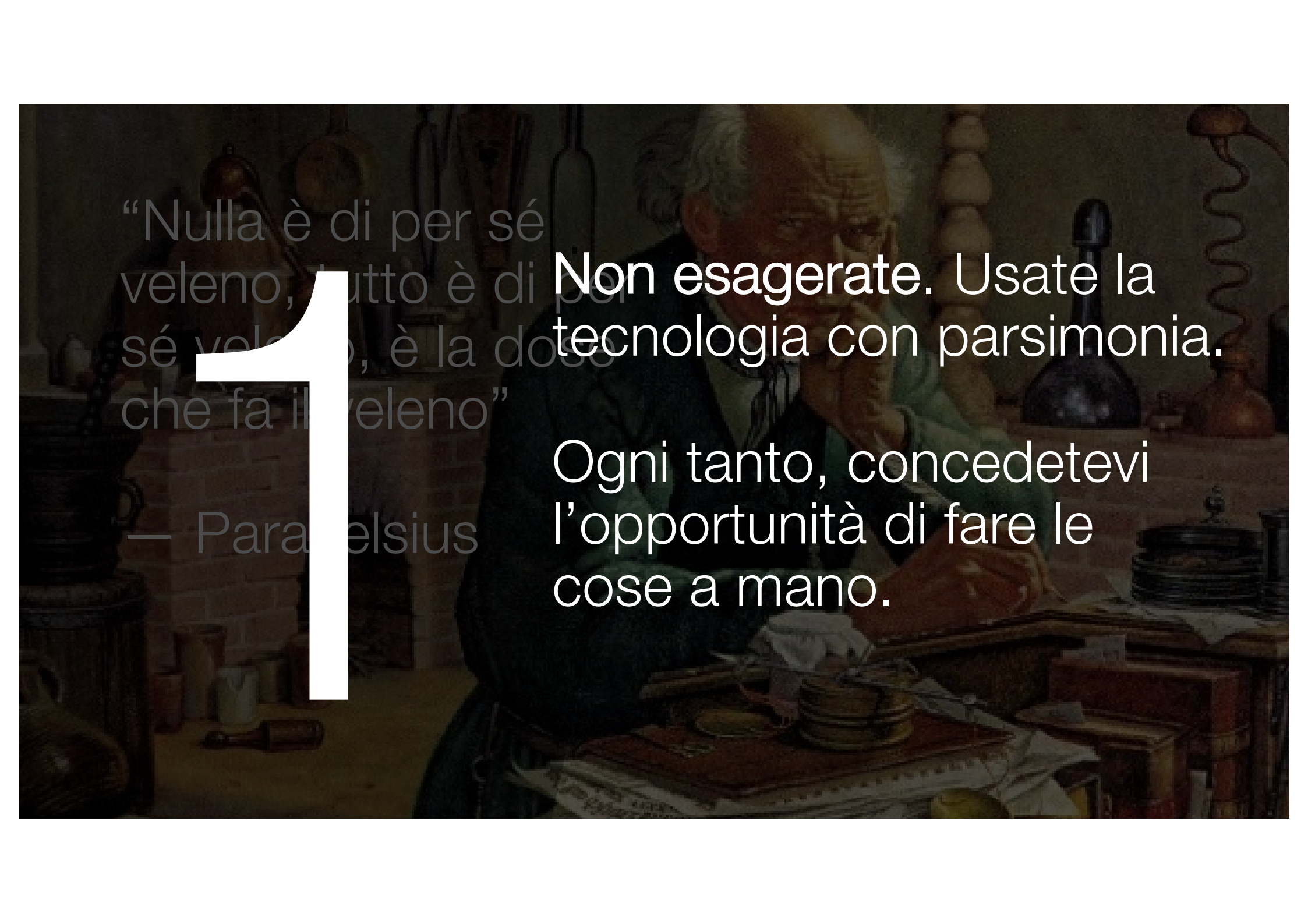
A close-up shot of a hand holding a dagger. The hand is dark-skinned and has a weathered, textured appearance. The dagger has a silver-colored metal blade with a sharp point. The handle is made of a dark, possibly black, material and is intricately carved with a complex, swirling pattern. Several large, oval-shaped red gemstones are set into the handle, adding to its ornate design. The background is a soft, out-of-focus brown color.

EQUILIBRIO

“Nulla è di per sé
veleno, tutto è di per
sé veleno, è la dose
che fa il veleno”

— Paracelsus



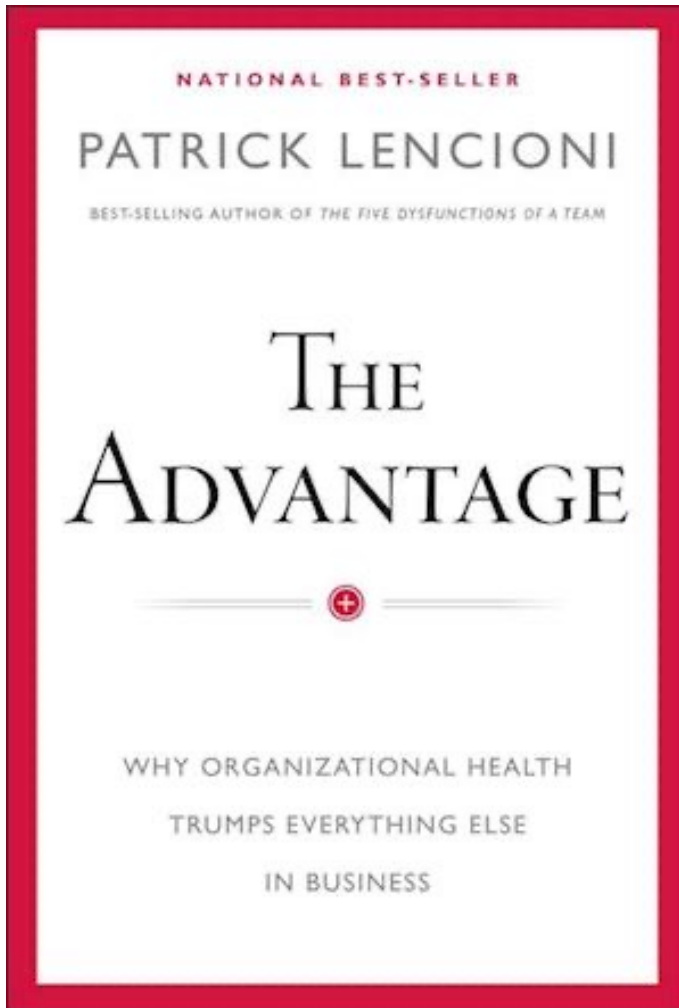


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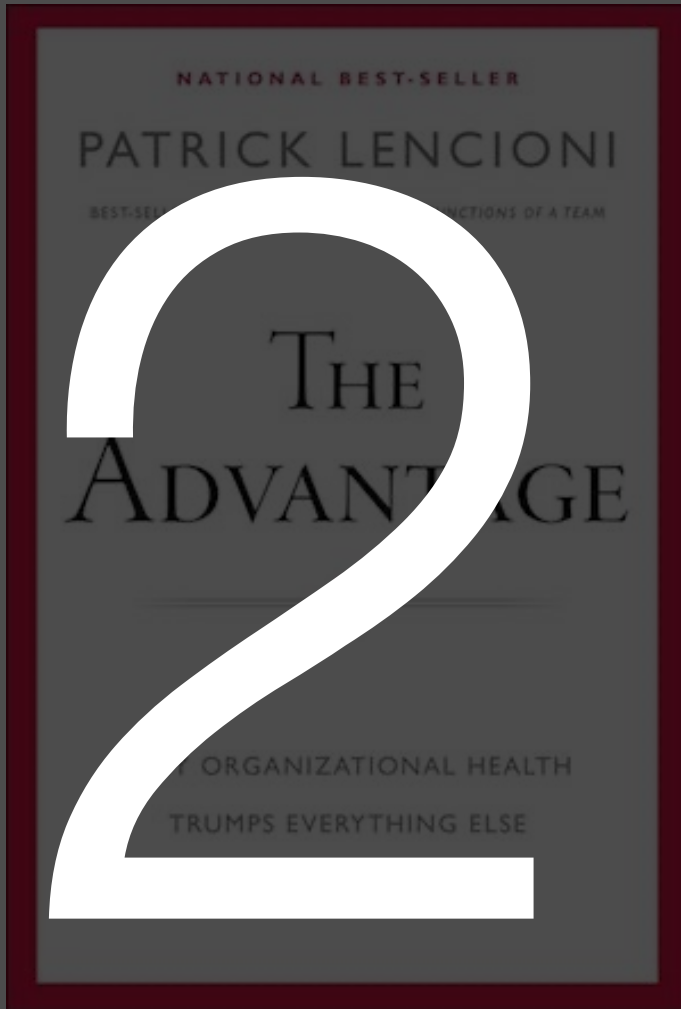
— Paracelsus

Non esagerate. Usate la
tecnologia con parsimonia.

Ogni tanto, concedetevi
l'opportunità di fare le
cose a mano.



“Quando si presenta un problema, prima di rivolgersi al capo, provate a trovare almeno due possibili soluzioni.”



Esercitatevi a pensare a soluzioni da soli prima di usare l'AI.

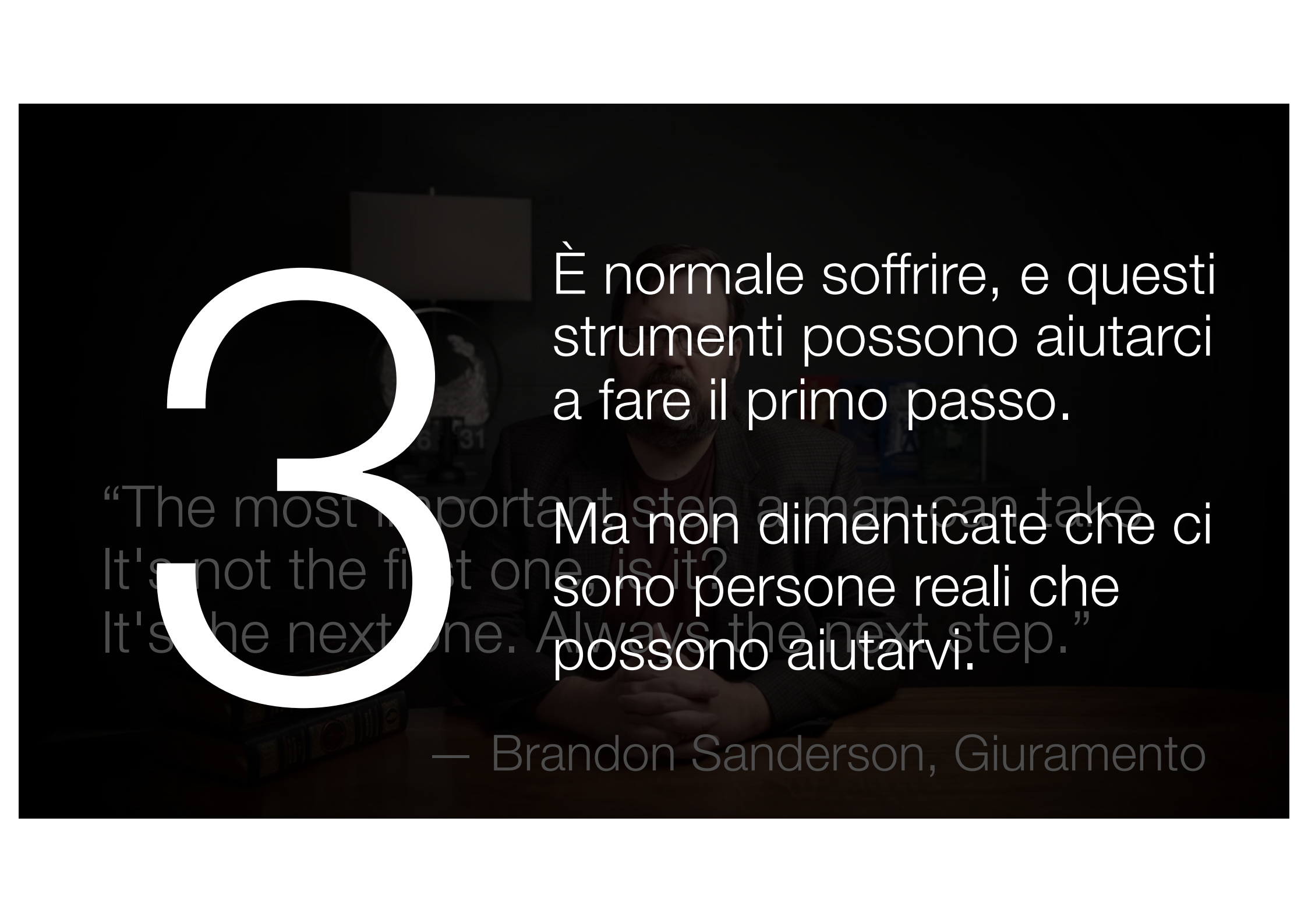
Quando si presenta un problema, prima di rivolgersi al capo, provate a trovare almeno due possibili soluzioni.

Non fatevi ingannare; tutti ci accorgiamo che quello che avete scritto è stato generato da una macchina.

A man with a beard and glasses, identified as Brandon Sanderson, is seated at a wooden desk. He is wearing a dark blazer over a maroon shirt. His hands are clasped in front of him. The background is dark, featuring a lamp with a white shade and a circular object, possibly a clock or a decorative item, on a shelf. To the right, there are some books or papers. In the foreground, on the left side of the desk, there are two thick books with gold lettering on their spines.

“The most important step a man can take.
It's not the first one, is it?
It's the next one. Always the next step.”

— Brandon Sanderson, Oathbringer



3

È normale soffrire, e questi strumenti possono aiutarci a fare il primo passo.

Ma non dimenticate che ci sono persone reali che possono aiutarvi.

— Brandon Sanderson, Giuramento





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The Double-Edged Sword of Digital Technologies

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rendere la
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1

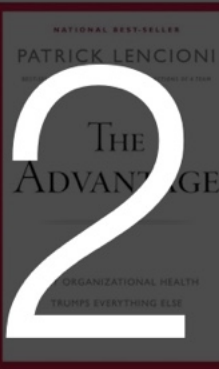
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2



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3

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strumenti possono aiutarci
a fare il primo passo.

Ma non dimenticate che ci
sono persone reali che
possono aiutarvi.

— Brandon Sanderson, Giuramento